

BAHIR DAR UNIVERSITY

BAHIR DAR INSTITUTE OF TECHNOLOGY

FACULTY OF MECHANICAL AND INDUSTRIAL ENGINEERING

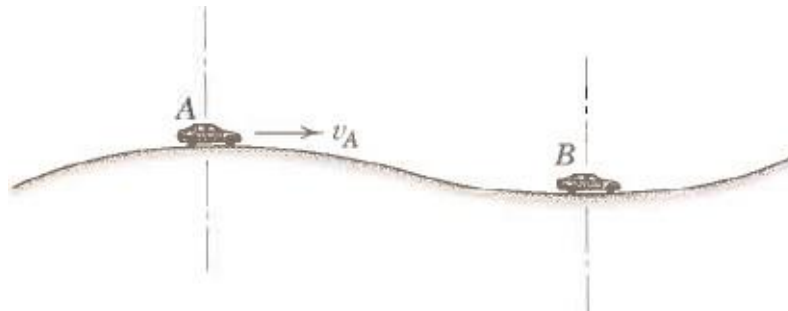
Engineering Mechanics II (Dynamics)

- Question number 1,2,4 and 8 are group assignment (In a group of maximum 5 students).
- Submission date: September 30, 2024.

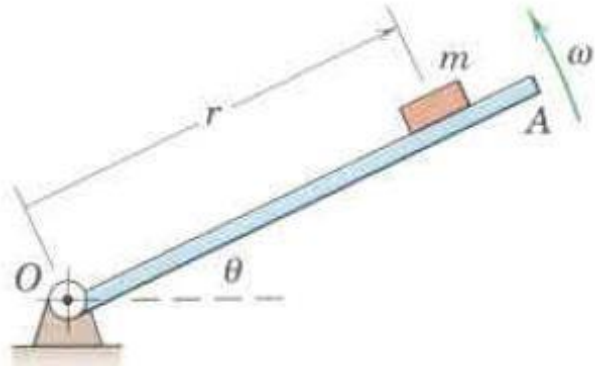
1. The 50 Kg crate is stationary when the force P is applied. Determine the resulting acceleration of the crate if, (a) $P=0$; (b) $P=150\text{N}$; and (c) $P=300\text{N}$.



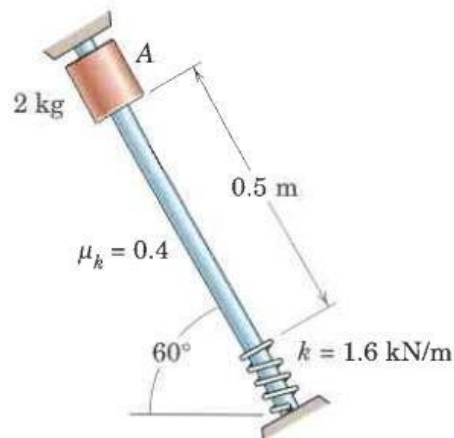
2. The car passes over the top of a vertical curve at A with a speed of 60 Km/hr and passes through the bottom of a dip at B. The radii of curvature of the road at A and B are both 100m. Find the speed of the car at B if the normal force between the road and the tires at B is twice that of at A. The mass center of the car is 1 meter from the road.



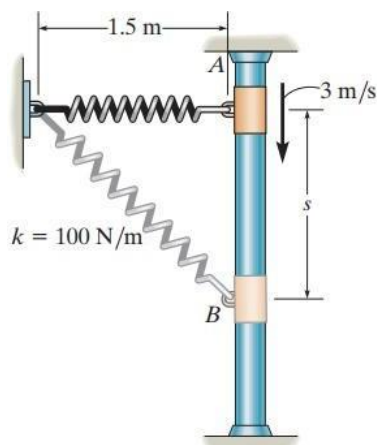
3. The member OA rotates about the horizontal axis through O with a constant counter clockwise angular velocity of $\omega = 3\text{rad/s}$. As it passes the position $\theta = 0$, a small block of mass m placed on it at radial distance $r = 0.46\text{m}$. If the block is observed to slip at $\theta = 50^\circ$, determine the coefficient of static friction between the block and the member.



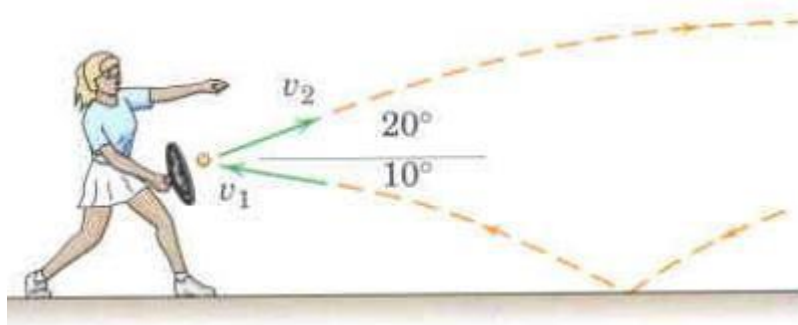
4. The 2-kg collar released from rest at A and slides down the inclined fixed rod in the vertical plane. The coefficient of kinetic friction is 0.4. Calculate (a) the velocity v of the collar as it strikes the spring and (b) the maximum deflection x of the spring.



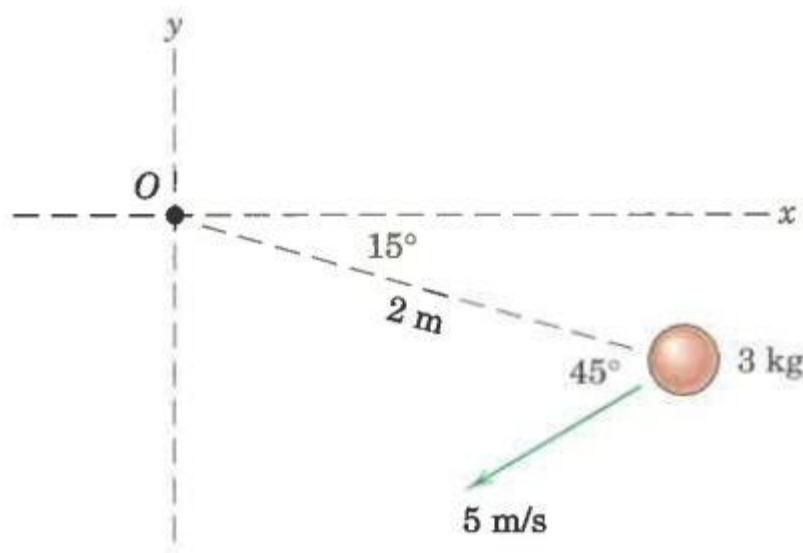
5. The 4-kg smooth collar has a speed of 3 m/s when it is at $s = 0$. Determine the maximum distance s it travels before it stops momentarily. The spring has an unstretched length of 1 m.



6. A tennis strikes the tennis ball with her racket while the ball is still rising. The ball speed before impact with the racket is $v_1 = 15\text{ m/s}$ and after impact its speed is $v_2 = 22\text{ m/s}$, which direction as shown in the figure. If the 60-g ball is in contact with the racket for 0.05 s, determine the magnitude of the average force R exerted by the racket on the ball. Find the angle β made by R with the horizontal. Comment on the treatment of the ball weight during impact.



7. The 3-kg sphere moves in the x-y plane and has the indicated velocity at a particular instant. Determine its (a) linear momentum, (b) angular momentum about point O, and (c) kinetic energy.



8. Two smooth disks A and B have the initial velocities shown just before they collide. If they have masses $m_A = 4\text{ kg}$ and $m_B = 2\text{ kg}$, determine their speeds just after impact. The coefficient of restitution is $e = 0.8$.

